

HR Employee Attrition Analytics Dashboard

Abstract

This project presents an interactive HR Employee Attrition Analytics Dashboard built in Power BI using the IBM HR Employee Attrition dataset. It provides insights into employee attrition, workforce demographics, salary distribution, overtime, and job satisfaction through interactive visualizations and KPIs.

Introduction

Employee attrition affects productivity, recruitment cost, and organizational growth. This dashboard converts HR data into meaningful insights that support better workforce planning.

Problem Statement

The objective is to identify departments, job roles, and employee characteristics associated with higher attrition using interactive business intelligence dashboards.

Objectives

- Analyze attrition by department and job role.
- Study overtime impact.
- Compare monthly income.
- Evaluate job satisfaction.
- Support HR decision-making.

Dataset Description

Dataset Overview: IBM HR Employee Attrition Dataset.

Features Used: Attrition, Department, Job Role, Age, Monthly Income, OverTime, Job Satisfaction, Gender, Marital Status, Education and related HR fields.

Number of Records and Columns: 1,470 records and 35 columns.

Tools & Technologies

- Power BI Desktop

- Power Query
- DAX
- IBM HR Employee Attrition Dataset

Data Preparation

Data Cleaning:

- Checked missing values and duplicates.
- Verified data types.

Power Query Transformations:

- Cleaned and formatted data.
- Prepared data model.

DAX Measures Created:

- Total Employees
- Attrition Count
- Attrition Rate
- Average Age
- Average Monthly Income
- Average Job Satisfaction

Dashboard Overview

Insert a full-page screenshot of your Power BI dashboard here.

The dashboard contains KPI cards, interactive charts, and slicers for Department, Gender, OverTime, and Marital Status.

KPI Explanation

Total Employees: Displays total workforce.

Attrition Count: Shows employees who left.

Attrition Rate: Percentage of employees who left.

Average Age: Average employee age.

Average Monthly Income: Average employee salary.

Average Job Satisfaction: Average satisfaction score.

Chart Explanation

Attrition by Department: Compares attrition across departments.

Attrition by Job Role: Identifies high-turnover roles.

Attrition by OverTime: Shows overtime impact on attrition.

Job Satisfaction Distribution: Displays employee satisfaction.

Average Monthly Income by Job Role: Compares salaries by role.

Key Business Insights

- Research & Development has the highest attrition.
- Overtime employees are more likely to leave.
- Managers earn the highest average monthly income.
- Job satisfaction influences employee retention.

Conclusion

This dashboard provides an interactive overview of HR metrics and supports data-driven decisions through clear visualizations and KPIs.

Future Scope

- Predict employee attrition using Machine Learning.
- Connect to live HR databases.
- Add historical trend analysis.
- Create executive-level dashboards.